A Toy-2D Study of Diffusion Posterior Sampling and Flow with Interpolant Guidance for Linear Inverse Problems

Tristan Montalbetti
Samuel Lhayani

March 2026

Abstract

This report presents a complete toy-2D benchmark for comparing Diffusion Posterior Sampling (DPS), plain Flow with Interpolant Guidance (FIG), and FIG+ on linear inverse problems. The study extends the original FIG GitHub repository, which primarily targets image restoration, with a low-dimensional experimental pipeline designed for direct geometric inspection of the posterior. In the final version of the benchmark, the diffusion prior is modeled by a residual, time-conditioned DDPM denoiser trained from scratch with exponential moving average (EMA) weights. This improved prior substantially strengthens the unconditional fit on both Two-Moons and Eight-Gaussians and makes the conditional comparison more informative. Across the reported experiments, FIG+ is consistently the strongest method on posterior-quality metrics, while plain FIG remains unstable and DPS becomes sensitive to guidance strength in the most constrained regimes. The toy setting remains the main asset of the study: one can visualize the prior manifold, the observation line, the reference posterior, and the full denoising trajectories of the competing samplers.

1 Introduction

Diffusion-based priors have become a powerful tool for inverse problems, but their behavior is often difficult to interpret in image space. Conditional samplers can fail for several intertwined reasons: the prior may be underfit, the sampler may be too deterministic, the conditioning gradient may be mis-scaled, or the target posterior may simply be highly multimodal. This motivates a deliberately low-dimensional benchmark where every element of the problem is visible.

The present work was developed in the context of the original *Flow with Interpolant Guidance* (FIG) repository. That repository implements diffusion and flow-based inverse-problem solvers for image tasks such as super-resolution, deblurring, and inpainting. Our goal here is complementary: rather than reproducing the full image pipeline, we construct a self-contained 2D environment in which the geometry of DPS, FIG, and FIG+ can be inspected directly.

The resulting benchmark answers three questions:

1. Can a compact DDPM prior trained from scratch recover non-Gaussian toy distributions such as Two-Moons and Eight-Gaussians?
2. How do DPS, plain FIG, and FIG+ behave under a one-dimensional noisy linear observation?
3. How much of the final behavior comes from the prior itself, and how much comes from the conditional solver?

2 The FIG Method: Theory, Likelihood, and Core Assumptions

In linear inverse problems, the goal is to recover a true signal x^* from a degraded measurement defined as:

$$y = Ax^* + n \tag{1}$$

where $A \in \mathbb{R}^{m \times d}$ is the forward measurement operator and $n \sim \mathcal{N}(0, \sigma_n^2 I)$ is independent measurement noise.

To solve this using a generative prior (like a diffusion or flow matching model), we cannot just sample blindly from the model; we must sample from the posterior distribution conditioned on our specific observation, y. This requires guiding the reverse-time generation process using the conditional score, $\nabla_{x_t} \log p_t(x_t|y)$.

2.1 Deriving the Guidance Term

To understand how this conditioning is applied, we decompose the conditional score using Bayes' theorem:

$$p_t(x_t|y) = \frac{p_t(x_t)q(y|x_t)}{p(y)} \tag{2}$$

Taking the logarithm of both sides and then the gradient with respect to the intermediate noisy state x_t yields:

$$\nabla_{x_t} \log p_t(x_t|y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log q(y|x_t) - \nabla_{x_t} \log p(y) \tag{3}$$

Because the marginal probability of the measurement $p(y)$ does not depend on the dynamic state x_t, its gradient is zero. The conditional score naturally splits into two distinct components:

$$\nabla_{x_t} \log p_t(x_t|y) = \underbrace{\nabla_{x_t} \log p_t(x_t)}_{\text{Unconditional Prior Score}} + \underbrace{\nabla_{x_t} \log q(y|x_t)}_{\text{Likelihood Score (Guidance)}} \tag{4}$$

2.2 Overcoming Intractability with the FIG Solution

As established, the conditional score naturally splits into an unconditional prior score and a likelihood guidance score. The first term is simply the unconditional score, which is readily provided by the pre-trained neural network at any timestep t.

The fundamental challenge in solving inverse problems lies in the second term: the likelihood score $\nabla_{x_t} \log q(y|x_t)$ is mathematically intractable. The static measurement y depends strictly on the fully clean, original image x^*, not the highly noisy intermediate state x_t. To evaluate this likelihood exactly, one would need to marginalize over all possible clean images that could have resulted in the current noisy state x_t:

$$q(y|x_t) = \int p(y|x_0)p(x_0|x_t)dx_0 \tag{5}$$

Evaluating this integral is computationally impossible, leaving standard generative models struggling to guide the process accurately without introducing significant approximation errors.

To bypass this intractability, the Flow with Interpolant Guidance (FIG) method proposes interpolating the measurement itself. Instead of attempting to condition the noisy state x_t on the static clean measurement y, FIG constructs a dynamic measurement interpolant y_t parallel to the forward process, using the same noise schedule coefficients (α_t, σ_t):

$$y_t = \alpha_t y_0 + \sigma_t y_T \tag{6}$$

By substituting the initial measurement $y_0 = Ax^* + n$ and the terminal noise $y_T = A\epsilon_x$ (where ϵ_x is the standard Gaussian noise used to generate x_T), this expands to:

$$\begin{aligned} y_t &= A(\alpha_t x^* + \sigma_t \epsilon_x) + \alpha_t n \\ y_t &= Ax_t + \alpha_t n \end{aligned} \tag{7}$$

Because the dynamic measurement y_t is strictly equal to the noisy image x_t passed through the operator A, plus scaled Gaussian noise, the intractable integral vanishes. The conditional likelihood becomes a perfectly known, exact Gaussian distribution:

$$q_t(y_t|x_t) = \mathcal{N}(y_t; Ax_t, \alpha_t^2 \sigma_n^2 I) \tag{8}$$

This allows the guidance gradient to be computed analytically as an L_2 norm, cleanly solving the intractability problem.

2.3 The Foundational Assumption

To mathematically justify this dynamic conditioning, the authors assume: *The conditional distribution of x_t given y_0 and ϵ_y is equivalent to that of x_t given y_t.*

- **Theoretical Ideal (Independence):** This implies the measurement noise (ϵ_y) and image noise (ϵ_x) are completely independent. This assumption elegantly removes boundary dependencies, making the conditional probability flow mathematically tractable.
- **Practical Reality (Dependence):** To maximize numerical efficiency, the implementation explicitly defines $\epsilon_y = A\epsilon_x$. This forcibly links the two noise variables, intentionally violating the theoretical assumption in the code.

3 Transitioning to Diffusion Models: The Backward SDE

While FIG is natively introduced using continuous normalizing flows, it can be adapted to diffusion models. Diffusion models define the forward data perturbation using a Stochastic Differential Equation (SDE) rather than strict interpolation coefficients.

To transition, the flow-matching interpolation coefficients must be mapped to the diffusion variance schedule $\beta(t)$. For a Variance-Preserving (VP) SDE, defined by $dx = -\frac{1}{2}\beta(t)xdt + \sqrt{\beta(t)}dW$, the equivalent coefficients that yield the exact same marginal distributions are:

$$\alpha_t = \sqrt{1 - e^{-\int_0^t \beta(s)ds}}, \quad \sigma_t = e^{-\frac{1}{2}\int_0^t \beta(s)ds} \tag{9}$$

With these defined, FIG can construct the measurement interpolant y_t exactly as before. To sample the posterior, we must derive the reverse-time conditional SDE. This derivation relies on Anderson's theorem and Bayes' rule:

1. **The Unconditional Base:** According to Anderson's theorem, for any forward diffusion SDE $dx = f(x,t)dt + g(t)dW$, the unconditional reverse-time generative process is:

$$dx = [f(x,t) - g(t)^2 s_t(x)]dt + g(t)d\bar{W} \tag{10}$$

   where $\bar{W}$ is the reverse-time Wiener process.

2. **Bayesian Decomposition:** When conditioning this reverse process on the measurement interpolant y_t, the unconditional score $s_t(x)$ is replaced by the conditional score $s_t(x|y_t)$. According to Bayes' rule, this decomposes linearly:

$$s_t(x|y_t) = s_t(x) + \nabla_x \log q_t(y_t|x) \tag{11}$$

3. **Substituting the Tractable Likelihood:** Because the FIG method guarantees that $q_t(y_t|x_t)$ is the Gaussian $\mathcal{N}(y_t; Ax_t, \alpha_t^2\sigma_n^2 I)$, the gradient of its log-likelihood is analytically exactly:

$$\nabla_x \log q_t(y_t|x) = -\frac{1}{2\alpha_t^2\sigma_n^2}\nabla_x ||y_t - Ax||_2^2 \tag{12}$$

Final Synthesis: By substituting this explicit guidance term back into the unconditional reverse SDE, and using a forward-running clock $\tau = T - t$ (where $\tau \in [0, T]$), we obtain the rigorous backward diffusion equation required to sample from the conditional posterior:

$$dx_\tau = \left[-f(x_\tau, T-\tau) + g(T-\tau)^2\left(s_{T-\tau}(x_\tau) - \frac{1}{2\alpha_{T-\tau}^2\sigma_n^2}\nabla_{x_\tau}||y_{T-\tau} - Ax_\tau||_2^2\right)\right]d\tau + g(T-\tau)d\bar{W}_\tau \tag{13}$$

4 Numerical Integration: Euler-Maruyama and Operator Splitting

Evaluating the unconditional score and the conditional guidance gradient simultaneously at every continuous time increment is both computationally prohibitive and numerically unstable. To resolve this, we employ the **Euler-Maruyama method** (or standard Euler for the Probability Flow ODE) combined with **operator splitting**.

4.1 The Necessity of Operator Splitting

Operator splitting decouples the reverse differential equation into two distinct physical processes: moving along the learned data manifold (the prior) and enforcing data consistency (the likelihood). This decoupling provides two major advantages:

- **Computational Efficiency:** Evaluating the score function requires a full forward pass through a massive neural network. Conversely, the likelihood gradient is a cheap linear algebra operation.
- **Numerical Stability:** The likelihood gradient around the measurement subspace can be extremely steep compared to the smooth unconditional prior.

Splitting allows us to query the expensive neural network only once per macroscopic timestep Δt, while safely taking multiple finer steps along the steep likelihood gradient to avoid overshooting.

4.2 The Two-Phase Algorithmic Step

The algorithm applies these decoupled terms sequentially at each discrete timestep t_i:

1. **Phase 1: Unconditional Update.** The pre-trained diffusion score network takes a step toward realistic data, ignoring the measurement. Using the PF-ODE formulation:

$$x'_{i-1} = x_i - \left[f(x_i, t_i) - \frac{1}{2} g(t_i)^2 s_\theta(x_i, t_i) \right] \Delta t \tag{14}$$

2. **Phase 2: Conditional Guidance Step.** To prevent the exact Gaussian guidance from overwhelming the prior, the algorithm calculates the measurement interpolant y_{i-1} and takes K iterative gradient descent steps to pull the state toward the measurement:

$$x'_{i-1} \leftarrow x'_{i-1} - \eta \nabla_x ||y_{i-1} - A x'_{i-1}||_2^2 \tag{15}$$

   where η is the task-specific learning rate. After K updates, the resulting tensor becomes x_{i-1} for the next iteration.

4.3 Tuning the K Iterations: The Consistency-Prior Trade-off

The task-specific hyperparameter K dictates the strength of data consistency, balancing the unconditional prior against the measurement information.

- **Underfitting (Small K):** Insufficient updates (e.g., $K = 1$) under-optimize consistency, yielding blurry reconstructions with unnatural details.
- **Overfitting (Large K):** Fully minimizing $||y_t - Ax_t||_2^2$ overpowers the prior and pushes the state off the data manifold, introducing severe noise and artifacts.
- **Optimal K:** Typically tuned via grid search alongside the learning rate η, this value maintains the critical balance between structural integrity and measurement fidelity.

5 The FIG+ Extension: Overcoming Information Sparsity in Inpainting

While standard FIG excels at global degradations (like super-resolution or deblurring), it struggles significantly with severe inpainting tasks where a large percentage of the image is missing.

5.1 The Problem

In inpainting, the forward operator A is a binary mask. For the heavily masked regions, the measurement interpolant y_t contains no structural information. Consequently, the gradient of the L_2 norm $\nabla_x ||y_t - Ax_t||_2^2$ is exactly zero in these masked regions. The conditional update fails because there is no signal to guide the unconditional prior.

5.2 The FIG+ Solution and Tweedie's Formula

To fix this, the authors introduced FIG+, which utilizes **Tweedie's formula** to actively hallucinate and inject missing information during the reverse process.

Tweedie's formula states that the score function implicitly contains the expectation of the clean data x_0 given the current noisy state x_t:

$$s_t(x_t) = -\alpha_t^{-1} \mathbb{E}(x_0 | x_t) \tag{16}$$

By rearranging this, we can formulate an explicit estimate of the clean image, $\hat{x}_0$, at any timestep t. In the context of rectified flow matching ($\alpha_t = t, \sigma_t = 1 - t$), this estimation translates mathematically to:

$$\hat{x}_0 = (1 - t) v_\theta(x_t, t) + x_t \tag{17}$$

5.3 Implementation of FIG+

At each discrete timestep, FIG+ implements three additional, computationally inexpensive matrix operations:

1. **Estimate Clean Data:** Use Tweedie's formula to estimate $\hat{x}_0$ from the current noisy state x_t.
2. **Forward Perturbation:** Re-add noise to $\hat{x}_0$ to project it back to the current timestep t_{i-1}, creating a synthetic target $x'_{i-1} = \alpha_{i-1}\hat{x}_0 + \sigma_{i-1}\epsilon_x$.
3. **Masked Mixing:** Combine the standard updated state x_{i-1} with the synthetic target x'_{i-1} exclusively in the masked regions (defined by $I - A$). A hyperparameter mixing weight $m \in [0, 1]$ controls how much to trust the hallucinated Tweedie estimate:

$$x_{i-1} = Ax_{i-1} + (1 - m)(I - A)x_{i-1} + m(I - A)x'_{i-1} \tag{18}$$

This effectively anchors the masked regions, preventing the generative process from drifting aimlessly.

6 Comparison: The Core Mechanisms of FIG vs. DPS

While both FIG and Diffusion Posterior Sampling (DPS) attempt to approximate the intractable conditional score, they rely on fundamentally different mathematical mechanisms for their primary guidance gradient. It is crucial to distinguish how each method utilizes (or avoids) clean data estimation.

6.1 Explicit Term Comparison and The Error Term

- **DPS** approximates the likelihood by mapping x_t to the clean data manifold using Tweedie's formula ($\hat{x}_0(x_t)$) and calculating the gradient of the distance to the static observation y_0. Its guidance term is:

$$g_{DPS} = -\rho\nabla_{x_t}||y_0 - A\hat{x}_0(x_t)||_2^2 \tag{19}$$

- **FIG** avoids $\hat{x}_0$ entirely in its gradient calculation. Instead of estimating clean data to compute the likelihood, it compares x_t directly to the noisy interpolant y_t. *(Note: Even in the FIG+ extension, Tweedie's formula is only used as a localized, post-hoc spatial mixing step—it is strictly excluded from the analytical gradient computation).* The core FIG guidance term is:

$$g_{FIG} = -\frac{1}{2\alpha_t^2\sigma_n^2}\nabla_{x_t}||y_t - Ax_t||_2^2 \tag{20}$$

The exact integral for the likelihood is $q(y|x_t) = \int q(y|x_0)p(x_0|x_t)dx_0$. DPS assumes $p(x_0|x_t)$ collapses perfectly into the expectation $\hat{x}_0$. Consequently, because DPS forces $\hat{x}_0$ inside its gradient, it introduces an approximation error (Δ_{error}) proportional to the ignored posterior variance:

$$\Delta_{error} \propto \mathrm{Var}(x_0|x_t) \cdot \nabla_{x_0}^2 \log p(y|x_0) \tag{21}$$

6.2 Justification of Methodological Efficacy

- **Compute Efficiency:** DPS requires calculating the Jacobian of the neural network ($\nabla_{x_t}\hat{x}_0(x_t)$) during the guidance step, meaning it must backpropagate through the entire U-Net architecture at every single sampling step. FIG computes its gradient analytically on the L_2 norm without any neural network backpropagation, making it significantly more computationally efficient per conditional update.
- **Noise Robustness:** At low noise levels ($t \to 0$), the generative variance $\mathrm{Var}(x_0|x_t)$ is small, making the DPS error term negligible. However, at high noise levels ($t \to T$), a single noisy state could correspond to thousands of clean images. DPS collapses this massive variance into a single guess, causing the Δ_{error} to explode and push the generation off the data manifold. FIG structurally respects this variance by matching noise distributions via the interpolant, making it vastly superior for high-noise and severely ill-posed scenarios.

7 Extension of the Original Repository

The upstream FIG repository is centered on image inverse problems with strong pretrained priors. It does not provide a dedicated 2D experimental environment. We therefore added the following components:

- synthetic data generators for `two_moons` and `eight_gaussians`;
- a DDPM training pipeline for 2D priors;
- a residual, time-conditioned MLP denoiser with EMA checkpointing;
- a linear inverse-problem module with Gaussian measurement noise;
- toy implementations of DPS, FIG, and FIG+;
- a reference posterior sampler based on importance reweighting of exact prior samples;
- evaluation scripts exporting CSV summaries, plots, and denoising animations.

This extension turns the original repository into a testbed where the conditioning mechanism can be examined independently of image-scale architectural complexity.

8 Problem Setup

8.1 Toy Distributions

We consider two standard non-Gaussian toy datasets:

- **Two-Moons**, a curved bimodal manifold that is useful for studying conditional sampling on nonlinear supports;
- **Eight-Gaussians**, a strongly multimodal distribution arranged on a ring, well suited to diagnosing mode preservation.

8.2 Improved Diffusion Prior

For each dataset, we train a DDPM prior from scratch. The forward process is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{22}$$

and the denoiser predicts $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ by minimizing

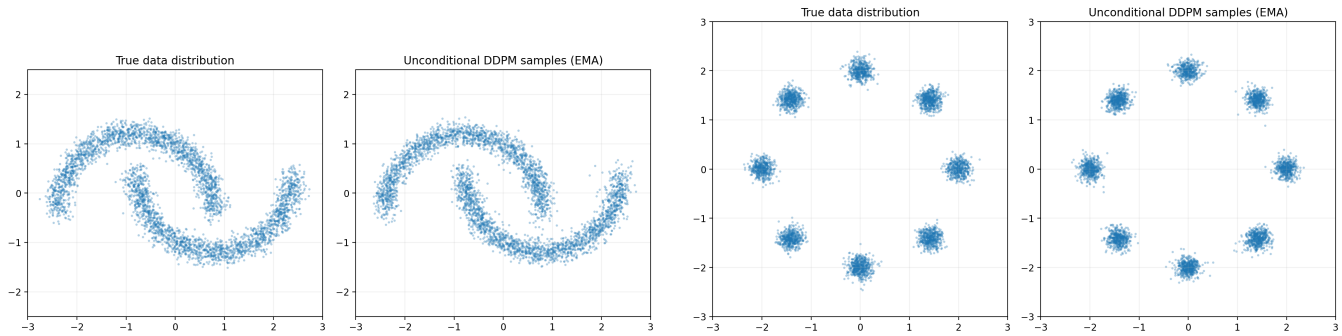
$$\mathcal{L}_{\mathrm{DDPM}} = \mathbb{E}_{\mathbf{x}_0,t,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right\|_2^2\right]. \tag{23}$$

The final denoiser architecture is a residual, time-conditioned MLP. This choice is intentional:

- the data lives in $\mathbb{R}^2$, so a U-Net would be unnecessary and would not exploit any meaningful spatial hierarchy;
- sinusoidal timestep embeddings provide a rich representation of the diffusion time;
- a small time MLP projects those embeddings into the hidden state dimension;
- each residual block applies LayerNorm, then a time-dependent scale-shift modulation, then a two-layer SiLU MLP;
- residual connections stabilize deeper networks and help preserve multimodal structure;
- an EMA of the weights is maintained during training and used for sampling.

This architecture is substantially stronger than the original plain MLP baseline. In practice, it solves two issues that were visible in earlier experiments: underfitting of Eight-Gaussians, and poor mode coverage under deterministic sampling.

The reported runs use 128 diffusion steps, a cosine schedule, 6000 optimization steps, batch size 1024, hidden dimension 256, time embedding dimension 128, 6 residual blocks, learning rate 3×10^{-4}, and EMA decay 0.995.



(a) Two-Moons prior sanity check.

(b) Eight-Gaussians prior sanity check.

Figure 1: Unconditional prior checks obtained with ancestral DDPM sampling and EMA weights. The improved denoiser recovers the global geometry of both toy distributions, which makes the subsequent conditional comparison more meaningful.

For unconditional sanity checks we use an ancestral DDPM sampler rather than a deterministic DDIM sampler. This matters in low dimension: DDIM can visually underrepresent multimodality even when the underlying denoiser is already good. For the inverse-problem solvers themselves, we keep the DDIM backbone so that DPS, FIG, and FIG+ are compared under the same deterministic reverse trajectory.

8.3 Linear Inverse Problem

We study the noisy linear observation model

$$\mathbf{y} = A\mathbf{x}^\star + \mathbf{n}, \qquad A = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 I). \tag{24}$$

Hence, the first coordinate is observed while the second coordinate remains hidden. This is the 2D analogue of a mask-like inpainting operator: the posterior is supported on the vertical line $x_1 = y$, but the distribution along that line remains structured because the prior is non-Gaussian.

8.4 Reference Posterior and Metrics

To evaluate the samplers against a meaningful target, we approximate the conditional posterior with importance reweighting. A large pool of exact prior samples $\{\mathbf{x}^{(i)}\}_{i=1}^N$ is drawn from the true toy distribution, and the weights

$$w_i \propto \exp\left(-\frac{\|A\mathbf{x}^{(i)} - \mathbf{y}\|_2^2}{2\sigma_n^2}\right) \tag{25}$$

define an empirical approximation of $p(\mathbf{x} \mid \mathbf{y})$.

We report:

- **Posterior Mean MSE**: error between the mean of generated samples and the mean of the reference posterior;
- **Sliced Wasserstein Distance (SWD)**: discrepancy between the generated and reference conditional distributions;
- **Measurement MSE**: average $\|A\mathbf{x} - \mathbf{y}\|_2^2$, which quantifies measurement consistency.

9 Conditional Solvers

9.1 Diffusion Posterior Sampling

DPS uses Tweedie's formula to estimate the clean sample

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}. \tag{26}$$

It then evaluates the measurement loss

$$\mathcal{L}_{\text{meas}}^{\text{DPS}} = \|\mathbf{y} - A\hat{\mathbf{x}}_0\|_2^2 \tag{27}$$

and backpropagates that loss to the current noisy state before taking the unconditional DDIM update.

9.2 Plain FIG

FIG introduces a measurement interpolant

$$\mathbf{y}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\mathbf{y} + w\sqrt{1-\bar{\alpha}_{t-1}}\,A\boldsymbol{\epsilon}, \tag{28}$$

with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$, then applies K gradient steps to the Gaussian negative log-likelihood

$$\mathcal{L}_{\text{meas}}^{\text{FIG}} = \frac{1}{2\sigma_n^2}\,\|\mathbf{y}_{t-1} - A\mathbf{x}\|_2^2\,. \tag{29}$$

In our implementation, the conditional step is scaled with a noise-dependent factor inspired by the SNR interpretation of FIG:

$$\eta_{t-1} = c\,\Delta t\,\frac{1-\bar{\alpha}_{t-1}}{\bar{\alpha}_{t-1}\,\|A\|_2^2}, \qquad \Delta t = \frac{1}{T}. \tag{30}$$

This provides a practical DDPM-compatible analogue of the theoretical FIG scaling.

9.3 FIG+ for Mask-Like Operators

Because the operator $A = [1, 0]$ acts like a coordinate mask, we also study FIG+, which mixes the hidden coordinate with a Tweedie-based estimate. After computing $\hat{\mathbf{x}}_0$, we perturb it back to the next time level,

$$\tilde{\mathbf{x}}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\boldsymbol{\epsilon}, \tag{31}$$

and combine it with the updated state through

$$\mathbf{x}_{t-1} \leftarrow P\mathbf{x}_{t-1} + (1-m)(I-P)\mathbf{x}_{t-1} + m(I-P)\tilde{\mathbf{x}}_{t-1}, \tag{32}$$

where $P = \text{diag}(1, 0)$ selects the observed coordinate and $m \in [0, 1]$ is a mixing coefficient. Intuitively, this keeps the observed coordinate measurement-consistent while restoring diversity in the hidden component.

10 Experimental Protocol

The evaluation is performed on both toy datasets with measurement noise levels

$$\sigma_n \in \{0.05, 0.5, 1.0\}.$$

For each dataset and each noise level, we use:

- 6 test observations;
- 128 conditional samples per solver and per observation;
- 128 reference posterior samples;
- a 20,000-sample importance pool for posterior approximation.

  For plain FIG and FIG+, we ablate

$$K \in \{1, 3, 5\}, \qquad w \in \{0.0, 0.5\}.$$

We keep the solver hyperparameters fixed across datasets: DPS uses $\zeta = 0.05$, FIG and FIG+ use learning rate 0.5, and FIG+ uses mixing coefficient $m = 0.95$.

11 Results

11.1 Quantitative Results on Two-Moons

Table 1: Two-Moons results with the improved residual DDPM prior. Lower is better for all metrics.

σ_n	Solver	Best (K, w)	Posterior Mean MSE	SWD	Measurement MSE
0.05	DPS	–	2.14×10^{-2}	0.314	2.40×10^{-3}
0.05	FIG	(5, 0.0)	8.86×10^{1}	51.804	5.93×10^{-3}
0.05	FIG+	(5, 0.5)	2.23×10^{-2}	0.212	5.65×10^{-3}
0.50	DPS	–	2.70×10^{1}	39.698	1.21
0.50	FIG	(5, 0.0)	4.01	38.656	1.11×10^{-1}
0.50	FIG+	(5, 0.5)	1.77×10^{-2}	0.314	2.29×10^{-2}
1.00	DPS	–	9.17	21.189	6.79×10^{-4}
1.00	FIG	(5, 0.0)	3.84	35.032	3.51×10^{-1}
1.00	FIG+	(5, 0.0)	1.90×10^{-2}	0.508	7.35×10^{-2}

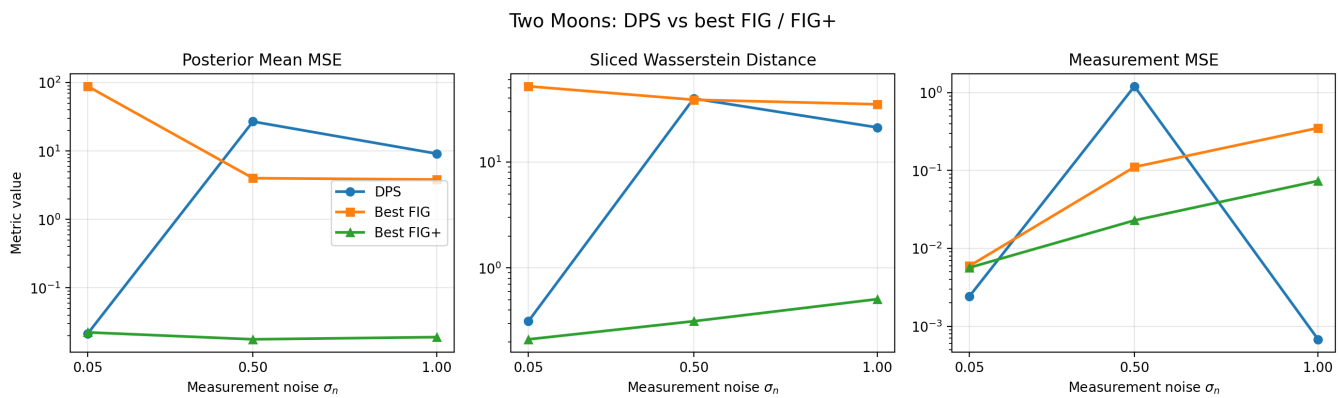


Figure 2: Two-Moons quantitative trends with the improved prior. FIG+ dominates the posterior-quality metrics across all tested noise levels. Plain FIG remains unstable, while DPS becomes highly sensitive to the guidance strength once the posterior becomes sharply concentrated.

With the stronger prior in place, the main bottleneck is no longer unconditional generation. The comparison now isolates the conditional solvers more clearly. On Two-Moons, FIG+ is the only method that stays close to the reference posterior across all noise levels. DPS is competitive at very low noise but becomes unstable at $\sigma_n = 0.5$ and $\sigma_n = 1.0$ under the fixed ζ used here, while plain FIG fails to preserve the bimodal posterior geometry.

11.2 Quantitative Results on Eight-Gaussians

Table 2: Eight-Gaussians results with the improved residual DDPM prior. Lower is better for all metrics.

σ_n	Solver	Best (K, w)	Posterior Mean MSE	SWD	Measurement MSE
0.05	DPS	–	1.51	10.209	9.57×10^{1}
0.05	FIG	(5, 0.5)	1.27×10^{2}	28.956	3.04×10^{-2}
0.05	FIG+	(5, 0.5)	1.91×10^{-2}	0.430	2.19×10^{-2}
0.50	DPS	–	5.85×10^{-2}	2.043	7.45×10^{-3}
0.50	FIG	(5, 0.0)	2.91	23.242	2.23×10^{-1}
0.50	FIG+	(5, 0.0)	1.17×10^{-2}	0.320	8.54×10^{-2}
1.00	DPS	–	1.11×10^{-1}	1.409	2.97×10^{-3}
1.00	FIG	(5, 0.0)	1.31	11.759	3.33×10^{-1}
1.00	FIG+	(5, 0.0)	4.21×10^{-2}	0.635	9.30×10^{-2}

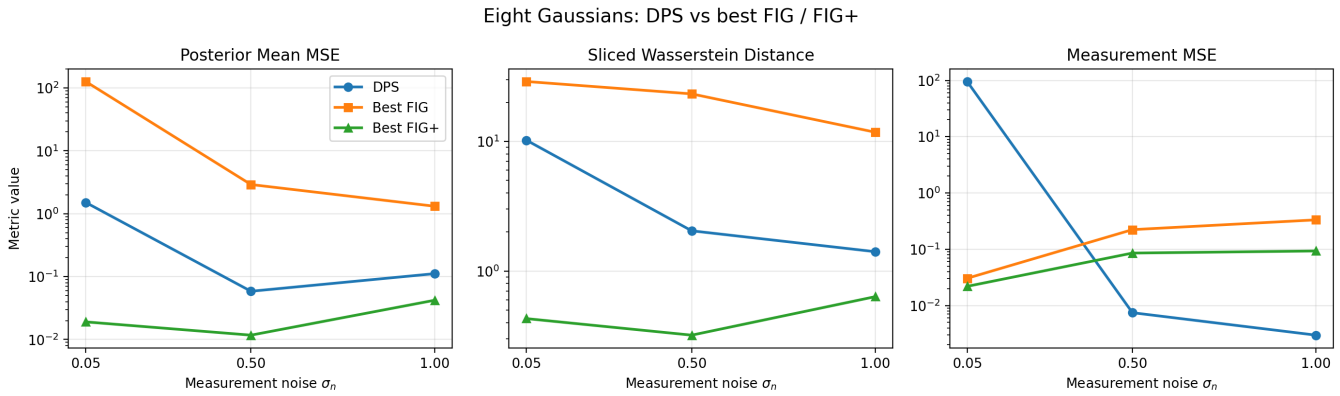
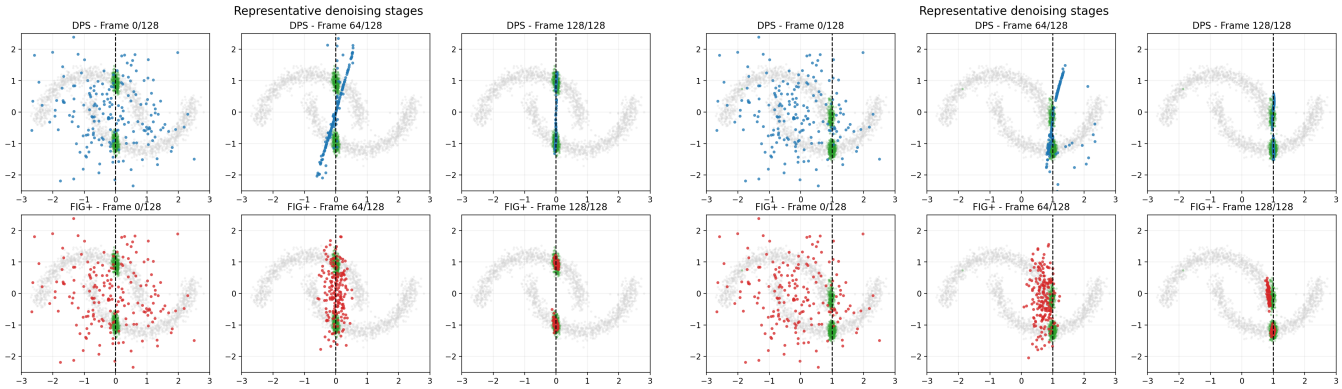


Figure 3: Eight-Gaussians quantitative trends with the improved prior. FIG+ again provides the best posterior fit. DPS remains competitive at moderate and high measurement noise, but deteriorates sharply at low measurement noise where the posterior is highly multimodal and tightly constrained.

Eight-Gaussians makes the mode-preservation issue even clearer. Because several clusters remain plausible under a fixed observed first coordinate, the solver must preserve multimodality rather than simply collapse toward a narrow line. FIG+ is consistently the strongest method of the three. DPS remains reasonable for $\sigma_n \in \{0.5, 1.0\}$, but becomes unreliable for $\sigma_n = 0.05$, where the posterior is both tight and multimodal.

11.3 Inpainting Dynamics Along Fixed Observation Lines

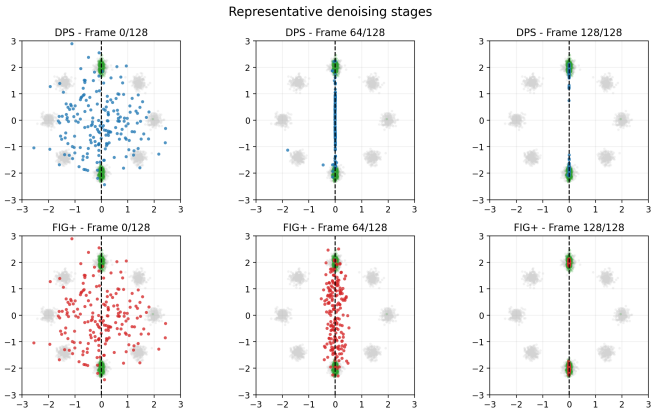
Static metrics only show the endpoint of the solver. To inspect the conditioning mechanism itself, we generated denoising animations from a shared Gaussian initialization for two fixed observation lines, $x_1 = 0$ and $x_1 = 1$, on both datasets. These four cases are particularly informative because they correspond to different posterior geometries: ambiguous symmetric conditioning at $x_1 = 0$, and more asymmetric conditioning at $x_1 = 1$.



(a) Two-Moons, fixed line $x_1 = 0$.

(b) Two-Moons, fixed line $x_1 = 1$.

Figure 4: Representative denoising stages for Two-Moons. Gray denotes the true prior support, green denotes the reference posterior, blue denotes DPS, and red denotes FIG+. FIG+ reaches the correct modes more directly and with less residual spread away from the observation line.



(a) Eight-Gaussians, fixed line $x_1 = 0$.

Figure 5: Representative denoising stages for Eight-Gaussians. The two visible posterior modes correspond to the two clusters compatible with the observed x_1. FIG+ remains concentrated around those modes, while DPS can overshoot or keep unnecessary spread before convergence.

These fixed-line experiments are especially helpful for understanding the inpainting interpretation of the task. The first coordinate is observed, the second is hidden, and the solver must reconstruct the distribution of plausible hidden coordinates conditioned on the observation. In this view, "landing on the line" is necessary but not sufficient: the real goal is to place mass at the correct modes *along* the line. The figures above show that FIG+ is much better at this than plain FIG and, in the present configuration, more stable than DPS.

The full animations are available alongside the paper at

```
   animations/two_moons_sigma_0.05_x1_p0.00_vs_dps_vs_fig_plus.gif
   animations/two_moons_sigma_0.05_x1_p1.00_vs_dps_vs_fig_plus.gif
animations/eight_gaussians_sigma_0.05_x1_p0.00_vs_dps_vs_fig_plus.gif
animations/eight_gaussians_sigma_0.05_x1_p1.00_vs_dps_vs_fig_plus.gif
```

12 Discussion

Several conclusions emerge from the final benchmark.

1. The prior architecture matters. The original plain MLP was too weak for Eight-Gaussians and could visually understate the true quality of the learned prior. The residual, time-conditioned denoiser with EMA changes that regime completely: the unconditional fit is now strong enough that the conditional results are primarily about the solver, not about prior failure.

2. FIG+ is the most reliable conditional sampler in this setting. Across both datasets and all tested noise levels, FIG+ achieves the best posterior mean MSE and the best SWD. This confirms that the Tweedie-style reinjection of the hidden coordinate is crucial when the observation behaves like a mask.

3. Plain FIG remains too brittle for the present operator. Even when its measurement MSE is not catastrophic, plain FIG often misses the posterior geometry. The issue is not only whether the line constraint is enforced, but whether the solver preserves the correct multimodal distribution along that line.

4. DPS is strong but hyperparameter-sensitive. With the improved prior, DPS remains competitive in some regimes, especially on Eight-Gaussians at moderate and high measurement noise. However, the fixed guidance strength used here is clearly not robust across all posterior geometries. This is visible in the sharp failure cases at Two-Moons $\sigma_n = 0.5$ and Eight-Gaussians $\sigma_n = 0.05$.

5. The toy-2D setting remains highly informative. The benchmark makes posterior geometry directly visible. One can see when a solver merely satisfies the measurement and when it actually reconstructs the correct conditional distribution. This interpretability is the main value of the toy benchmark.

13 Limitations and Future Work

The current paper focuses on a deliberately minimal setting. Several extensions would be natural:

- retuning DPS guidance strength as a function of the prior and measurement noise;
- larger evaluation sweeps with more test observations and more reference samples;
- additional linear operators, including rotated projections and full-rank noisy maps;
- nonlinear observation models;
- a direct bridge from the toy benchmark back to the image-scale operators of the original FIG repository.

These directions would help separate which phenomena are generic properties of the solver and which are specific to the present mask-like inverse problem.

14 Conclusion

This work transforms the original FIG repository into a complete toy-2D benchmark for inverse problems. The final version of the benchmark includes stronger residual DDPM priors, EMA sampling, refreshed quantitative evaluations, and fixed-line denoising animations that expose the conditional dynamics directly. With this improved setup, the conclusion is clear: FIG+ is the most effective method overall on posterior-quality metrics, plain FIG remains fragile, and DPS is competitive but sensitive to guidance tuning. More broadly, the study shows why toy benchmarks are valuable: they expose the geometry of conditional generation directly and make algorithmic behavior observable rather than merely measurable.